Made By| Adeel Shahbaz, Abdul Rehman,
Asad Mehmood

University Of Management & Technology

Study Buddy

Study Buddy

Project Overview

Study Buddy is an AI-powered educational platform developed using Streamlit and the OpenRouter API. The platform is designed to help students study more effectively by converting study materials into interactive learning resources. It supports different file formats such as PDF, TXT, MD, and PPTX.

The system automatically analyzes uploaded content and generates useful study tools including notes, flashcards, quizzes, and study plans. The main goal of the project is to improve learning, increase productivity, and help students understand topics in a more organized way.

System Architecture and Workflow

1. File Processing System

The application uses a lightweight text extraction system to read content from uploaded files. Once the text is extracted, it is stored securely in the session state.

This approach provides several benefits:

- Reduces repeated processing
- Improves application speed
- Makes the user interface more responsive
- Ensures smooth interaction while using the platform

---

2. AI-Based Content Generation

Study Buddy uses advanced Large Language Models (LLMs) through the OpenRouter API to generate learning materials automatically.

The system can create:

- Structured study notes
- Flashcards for active recall learning
- Multiple-choice quizzes
- Topic summaries

A controlled prompting system is used to ensure that all generated content is based only on the uploaded study material. This helps reduce incorrect or unrelated AI-generated information.

---

3. Smart Study Sequencing

The platform includes an intelligent sequencing feature that organizes topics in a logical order.

The system:

- Analyzes document summaries
- Identifies important concepts
- Arranges topics from basic to advanced levels

This helps students follow a clear learning path and understand concepts step by step.

Productivity Features

1. Task Manager

Study Buddy includes a built-in task management system that helps students organize their academic work efficiently.

Features include:

- Priority-based task organization
- Progress tracking
- Academic workload management

---

2. Pomodoro Focus Timer

The application also provides a Pomodoro focus timer based on the 25/5 productivity method:

- 25 minutes focused study
- 5 minutes break

This feature helps users maintain concentration and improve study efficiency.

Analytics and Progress Tracking

The platform records and analyzes user activity to provide real-time academic insights.

Tracked data includes:

- Quiz performance
- Study sessions
- Completed focus sessions
- Productivity statistics

The collected information is visualized using Plotly charts and graphs, allowing users to monitor their learning progress easily.

Key Features Summary

- AI-powered study material generation

- Support for multiple document formats
- Smart topic sequencing
- Interactive quizzes and flashcards
- Built-in productivity tools
- Real-time analytics and progress tracking
- Fast and responsive user experience

Technologies Used

- Python
- Streamlit
- OpenRouter API
- Large Language Models (LLMs)
- Plotly